Subject Description Form

Subject Code	ABCT5037
Subject Title	Green Chemistry for Sustainable Products Development
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims to introduce the principles and the application of green chemistry for developing eco-friendly and safe chemical products. Students will learn to identify major environmental challenges associated with chemical industries. Students will be equipped with tools available to scientists and engineers to practice green, energy-efficient and safe chemistry. This subject will also provide students with updated development of selected emerging technologies for sustainable chemical synthesis.
Intended Learning Outcomes	Upon the successful completion of this subject, students will be able to: a) identify common environmental sustainability issues associated with chemical processes; b) apply the concepts and principles of green chemistry to develop sustainable chemical processes and products; c) analyze the efficiencies and limitations of different green chemistry tools for designing sustainable processes and products; d) recognize some latest developments in new chemical processes and technologies to develop greener products.
Subject Synopsis/ Indicative Syllabus	**1. Green chemistry and processes in the context of sustainability**: • Environmental, health and safety issues • Principles of green chemistry for sustainable development **2. Strategies and principles for developing sustainable processes**: • Waste minimization and atom economy • Reduction of materials use (catalytic reactions; reduction of non-renewable feedstocks or starting materials) • Reduction of energy requirement (energy efficiency improvement; alternative energy sources)

	• Reduction of risk and hazard (safe product design; alternative solvents) **3. Measuring and controlling environmental performances** • Evaluating the effects of chemicals on human, wildlife and local environment • Introduction of Life Cycle Assessment (LCA) methodology and framework • Product- and process-oriented LCA • Evaluating methods to design safer chemicals **4. Catalysis and green chemistry** • Introduction to catalysis • Heterogeneous and homogeneous catalysts for bulk and fine chemical industries • Catalytic oxidation using hydrogen peroxide • Biocatalysis **5. Organic solvents** • Organic solvents and volatile organic compounds • Solvent-free systems • Supercritical CO_2 as the solvent • Ionic liquids and fluorous biphasic solvents • Comparing of green-ness of solvents **6. Renewable feedstocks and starting materials** • Chemicals from fatty acids • Polymers from renewable feedstocks • Conversion of biomass to some other chemicals • Catalysts from chemical wastes **7. Emerging green technologies for green chemical processes** • Design processes for energy and atom-efficiency • Photochemical reactions • Chemistry with microwaves • Sonochemistry • Electrochemical synthesis
Teaching/Learning Methodology	**Lecture**: basic concepts and working principles will be introduced and discussed with particular emphasis on the latest applications. Examples will be used to illustrate the applications of various methods and techniques. Tutorial sessions will be used to consolidate the contents of lectures and guide the students in problem solving and discussion.

	Presentation and written essay: The class will be divided into groups with 2~3 students per group, and various topics of the subject will be assigned to each group. Students will conduct literature research on the topic, present it in the class, and submit an individual written essay.
Assessment Methods in Alignment with Intended Learning Outcomes	(see table below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d		
1. Test	30%	✓	✓	✓	✓		
2. Presentation	30%	✓	✓	✓	✓		
3. Examination	40 %	✓	✓	✓	✓		
Total	100%						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Test: As a mean of timely checking of learning progress, in-class tests will be conducted during the semester to assess the students' understanding of the topic by referring to the intended learning outcomes.

Presentation: Students are required to prepare an oral presentation on assigned topics related to the emerging technologies for sustainable chemical synthesis. They will work in a small group for literature searching, discussion, and preparation of slides and a written report. Understanding of the topic, communication skill, and critical/creative thinking will be assessed.

Examination is a major assessment component of the subject, which will be conducted as a closed-book examination. This will be used to test the students' overall understanding of all intended learning outcomes and problem-solving skills.

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Presentation	6 Hrs.
	Other student study effort:	
	▪ Self-study and group work	81 Hrs.

	Total student study effort	120 Hrs.
Reading List and References	Anne E. Marteel-Parrish, Martin A. Abraham, Green Chemistry and Engineering: A pathway to sustainability, Wiley, 2014. Paul T. Anastas, John Charles Warner, Green Chemistry: Theory and Practice, Oxford University Press, 1998. Davor Margetic, Vjekoslav Štrukil, Mechanochemical Organic Synthesis, Elsevier, 2016 J. L. Luche, Synthetic Organic Sonochemistry, Plenum Press, 2001 Jean-Marc Lévêque, Giancarlo Cravotto, François Delattre, Pedro Cintas, Organic Sonochemistry: Challenges and Perspectives for the 21st Century, Springer, 2018 Georgios Stefanidis, Andrzej Stankiewicz, Alternative Energy Sources for Green Chemistry, RSC publication, 2016 Cheanyeh Cheng, Enzyme-Based Organic Synthesis, Wiley, 2022 Andreas Sebastian Bommarius, Bettina R. Riebel, Biocatalysis, Wiley, 2004	